

Foundation (Jalapeno)

- Q1.** Factor out the greatest common factor from $2x + 6$
(A) $2(x + 3)$
(B) $6(x + 2)$
(C) $2(x - 3)$
(D) $6(2x + 1)$
(E) $2(3x + 1)$
- Q2.** Simplify the expression $9x^2 + 12x$ by factoring
(A) $3x(3x + 4)$
(B) $3x(3x - 4)$
(C) $12x(9x + 1)$
(D) $3x(3x + 1)$
(E) $12x(3x - 1)$
- Q3.** What is the greatest common factor of 15 and 25?
(A) 1
(B) 5
(C) 10
(D) 15
(E) 20
- Q4.** Factor out the greatest common factor from $12x^2 + 16x$
(A) $4x(3x + 4)$
(B) $4x(3x - 4)$
(C) $12x(3x + 2)$
(D) $4x(3x + 1)$
(E) $12x(x + 1)$
- Q5.** Simplify the expression $20x + 16$ by factoring
(A) $4(5x + 4)$
(B) $4(5x - 4)$
(C) $20(x + 1)$
(D) $4(5x + 1)$
(E) $16(x + 1)$
- Q6.** A company has two types of stocks: A and B. Stock A has a 15
(A) 5
(B) 10
(C) 15
(D) 20
(E) 25
- Q7.** A tax consultant needs to calculate the tax on a client's savings. If the client has 1200 in savings with a 10
(A) 10
(B) 20
(C) 40
(D) 60
(E) 100
- Q8.** Simplify the expression $8x^2 + 12x$ by factoring
(A) $4x(2x + 3)$
(B) $4x(2x - 3)$
(C) $8x(x + 2)$
(D) $4x(2x + 1)$
(E) $4(2x^2 + 3x)$

Open Ended Questions (FRQ)

- Q9.** A stock market analyst wants to factor out the greatest common factor from the expression $18x^2 + 24x$. Simplify the expression and explain your reasoning.
- Q10.** A financial advisor needs to calculate the tax on a client's investment. If the client has 800 invested with a 15

Chef's Tips

- Make sure students understand the concept of greatest common factor
- Emphasize the importance of factoring in simplifying expressions
- Remind students to check their work and explain their reasoning